

Fine-Tuning Pretrained Transformer Models for Hate Speech Detection

M1 SDSI — 2026

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1. Introduction

This report presents the implementation and results of a transformer fine-tuning project for hate speech detection. The main objective is to understand the fine-tuning process for pretrained transformer models on a downstream NLP task and to evaluate their performance against a classical baseline.

Hate speech detection was selected as the target task, and the HateXplain dataset was used as the primary benchmark. The task is formulated as a three-class classification problem: hateful, offensive, and normal. Three transformer models were fine-tuned — DistilBERT, RoBERTa, and HateBERT — and a soft-voting ensemble was constructed as a bonus advanced experiment. A second bonus experiment extended the study to Croatian hate speech detection using multilingual BERT (mBERT).

1.1 Transformer Models

Transformer models are encoder architectures based on the self-attention mechanism, which allows each word to be represented in the context of all other words in the input sequence. This contextual representation is far richer than traditional bag-of-words or TF-IDF approaches, which treat words independently. Models such as BERT, RoBERTa, and DistilBERT are first pretrained on massive unlabeled text corpora to acquire broad linguistic knowledge.

1.2 Fine-Tuning

Fine-tuning is the process of continuing the training of a pretrained transformer on a task-specific labeled dataset. The model adapts its internal representations to the target classification objective while retaining its linguistic prior. This allows strong performance even on relatively small datasets, and is the central methodology demonstrated in this project.

1.3 Task and Dataset

Hate speech detection on HateXplain involves classifying social media posts into three categories. The offensive class occupies a semantic gray zone between hateful and normal content, making the task more representative of real-world detection challenges than simpler binary formulations.

2. Dataset and Preprocessing

2.1 HateXplain

HateXplain is a publicly available dataset for hate speech analysis hosted on Hugging Face (Hate-speech-CNERG/hatexplain). It contains 19,229 social media posts from Gab and Twitter, each annotated by three human annotators with one of three labels: hateful (0), normal (1), or offensive (2). The final label is determined by majority vote. The dataset provides official train/validation/test splits.

The class distribution in the training set is: normal (40.64%), hateful (30.87%), and offensive (28.50%). This moderate imbalance was handled through class-weighted loss functions in transformer experiments and class-weighted Logistic Regression in the baseline. Text lengths range from 2 to 165 words (median: 21 words), guiding the choice of max_length=128 tokens.

2.2 Preprocessing

Two preprocessing strategies were applied depending on the model type:

- **Transformer cleaning (light):** special placeholders normalized (<user> → USER, <number> → NUMBER), whitespace standardized. Semantic content preserved for contextual tokenization.

- **Baseline cleaning (strong):** lowercasing, placeholder normalization, removal of all non-alphanumeric characters, producing a clean surface form for TF-IDF feature extraction.

The official train/validation/test splits were preserved throughout all experiments.

3. Methodology

3.1 Baseline: TF-IDF + Logistic Regression

A TF-IDF vectorizer (max 20,000 features, unigrams + bigrams, min_df=2) combined with Logistic Regression was used as the classical baseline. Two variants were evaluated: standard and class-weighted (class_weight='balanced'). The balanced variant was selected as the final baseline due to its better macro F1-score.

3.2 DistilBERT

DistilBERT (distilbert-base-uncased) retains 97% of BERT's performance while being 40% smaller and 60% faster. Fine-tuned for 4 epochs with lr=2e-5, batch size=16, weight_decay=0.01, standard cross-entropy loss. Best model selected by validation macro F1.

3.3 RoBERTa (Class-Weighted)

RoBERTa (roberta-base) is a robustly optimized BERT variant. A WeightedTrainer with class-weighted cross-entropy loss was used (hateful: 1.0800, normal: 0.8203, offensive: 1.1696). Training: 4 epochs, lr=2e-5, batch size=16, 500 warmup steps.

3.4 HateBERT (Domain-Specific)

HateBERT (GroNLP/hateBERT) is a BERT-base model further pretrained on approximately 100 million hate-related Reddit posts from banned communities (the RAL-E dataset). Its domain-specific linguistic knowledge was hypothesized to give it an advantage on hate speech classification. The same WeightedTrainer configuration as RoBERTa was applied.

3.5 Soft-Voting Ensemble (Bonus)

The softmax probability distributions from DistilBERT, RoBERTa, and HateBERT were averaged and the argmax taken. This soft-voting approach uses confidence information from all models and is more robust than hard voting, particularly for borderline examples near class boundaries.

3.6 Multilingual Extension — Croatian FRENK (Bonus)

Multilingual BERT (bert-base-multilingual-cased) was fine-tuned on the FRENK Croatian dataset (classla/FRENK-hate-hr), a corpus of Facebook comments annotated as acceptable or unacceptable. The same training configuration and WeightedTrainer were used. A TF-IDF baseline using character n-grams (2–4 grams, char_wb) was also evaluated, which is more appropriate for morphologically rich languages like Croatian.

4. Results

4.1 English Hate Speech Detection (HateXplain)

Table 1 presents the test set results for all English models. M = macro average, W = weighted average, Off. F1 = offensive class F1-score.

Model	Accuracy	Prec(M)	Rec(M)	F1(M)	F1(W)	Off. F1
Baseline Balanced	0.6616	0.6517	0.6535	0.6525	0.6612	0.5193
DistilBERT	0.6959	0.6854	0.6870	0.6861	0.6953	0.5419
RoBERTa (Weighted)	0.6772	0.6675	0.6733	0.6681	0.6770	0.5199
HateBERT	0.6949	0.6830	0.6885	0.6848	0.6936	0.5421
Ensemble ★	0.7011	0.6895	0.6950	0.6911	0.7001	0.5488

Table 1 — Test set results on HateXplain. Best values (Ensemble) highlighted in green.

The results show a clear progression from baseline to ensemble. DistilBERT (+3.43% accuracy) and HateBERT (+3.33%) are the strongest single models. Notably, RoBERTa with class-weighted loss underperforms both despite being architecturally stronger, suggesting that weighted loss alone does not compensate without domain-specific pretraining. The ensemble achieves best results across all metrics: 70.11% accuracy and 0.6911 macro F1.

The offensive class remains the hardest category throughout, with F1 scores in the range 0.52–0.55. This is consistent with the known difficulty of this class in the literature, reflecting its semantic proximity to both hateful and normal content.

4.2 Cross-Language Comparison (Bonus)

Table 2 summarizes results across both languages.

Language	Model	Accuracy	F1 Macro
English (3-class)	TF-IDF Baseline	0.6616	0.6525
	DistilBERT	0.6959	0.6861
	RoBERTa (Weighted)	0.6772	0.6681
	HateBERT	0.6949	0.6848
	Ensemble	0.7011	0.6911
Croatian (binary)	TF-IDF Baseline	0.7607	0.7592
	mBERT	0.7513	0.7476

Table 2 — Cross-language results: English (3-class HateXplain) vs Croatian (binary FRENK).

mBERT achieves 75.13% accuracy and 0.7476 macro F1 on Croatian hate speech, substantially higher than the English results. This is largely explained by the task difference: Croatian is a binary classification (acceptable vs unacceptable), while English is a three-class problem with an ambiguous middle class. The character n-gram TF-IDF baseline is stronger on Croatian (0.7592 F1) than word-level baselines on English, confirming the suitability of character-level features for morphologically rich languages.

5. Discussion

5.1 What Worked Well

- All transformer models substantially outperformed the TF-IDF baseline, confirming the value of contextual representations for hate speech detection.
- The soft-voting ensemble achieved the best results across all metrics, demonstrating that model diversity reduces individual prediction errors.
- HateBERT's domain-specific pretraining provided a consistent advantage over general-purpose transformers, confirming the value of domain-targeted pretraining data.
- The class-weighted loss improved the offensive class F1 for RoBERTa, though at the cost of overall accuracy.
- The multilingual pipeline generalized effectively to Croatian without any language-specific adaptation beyond tokenization.

5.2 Advantages of Transformers for Hate Speech Detection

- Contextual representations allow transformers to distinguish hate speech from normal content even when similar words appear in different contexts.
- Transfer learning enables strong performance on limited labeled data by starting from a powerful linguistic prior.
- Domain-specific models like HateBERT demonstrate that targeted pretraining data significantly improves task performance.
- Multilingual models like mBERT generalize fine-tuning to non-English languages without language-specific architectures.

5.3 Challenges Faced

- The offensive class remained consistently the hardest category (F1 ~0.52–0.55), reflecting its semantic overlap with both hateful and normal posts.
- DistilBERT slightly outperformed RoBERTa despite being the smaller model, suggesting class-weighted loss was not fully effective without domain-specific pretraining.
- Training time was substantial (40–90 minutes per model on a T4 GPU). GPU memory constraints required careful batch size configuration.
- Arabic hate speech datasets on Hugging Face required login authentication, necessitating a pivot to the Croatian FRENK dataset for the multilingual experiment.

5.4 Potential Improvements

- Using larger models (BERT-large, RoBERTa-large, DeBERTa-v3) would likely push accuracy above 75%.
- Data augmentation for the offensive class (back-translation, synonym substitution) could reduce class confusion.
- Incorporating HateXplain's rationale annotations as auxiliary supervision signals could improve class boundary detection.
- Fine-tuning HateBERT with class-weighted loss may combine domain pretraining and imbalance handling.
- XLM-RoBERTa would provide stronger multilingual representations than mBERT for the Croatian experiment.

6. Conclusion

This project successfully implemented the complete fine-tuning pipeline for pretrained transformer models on a multiclass hate speech detection task. All assignment requirements were met: transformer theory was presented, a public dataset was selected and described, preprocessing was justified, three transformer models were fine-tuned with documented hyperparameters, results were evaluated using Accuracy, Precision, Recall, and F1-score, confusion matrices and learning curves were visualized, and a comparison with a classical baseline was provided.

All three bonus objectives were also achieved: three models were compared, the multilingual experiment extended the work to Croatian, and a soft-voting ensemble was implemented as the advanced technique. The ensemble achieved the best performance overall (70.11% accuracy, 0.6911 macro F1), confirming that combining complementary models reduces individual errors.

The central finding is that the offensive class represents a persistent and fundamental challenge that cannot be fully resolved by model scale alone. Domain-specific pretraining and class-aware training strategies both help, but the semantic overlap between classes remains the primary bottleneck. Future work should focus on richer supervision signals and larger domain-specific models.